Danny Muir

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EDUCATION

University of Washington, Seattle *Expected June 2027*
Bachelor of Science in Computer Engineering — **GPA: 3.84** Dean's List All Quarters 2023–2026
Paul G. Allen School of Computer Science & Engineering
Minor: Neural Computation and Engineering
Honors: Tau Beta Pi Engineering Honor Society; CSE Honors
Relevant Coursework: Machine Learning for Neuroscience (graduate), Machine Learning, Natural Language Processing, Neural Technology Studio, Neurocomputation/Engineering Lab, Scientific Computing, Digital Design (FPGA)

PUBLICATIONS

M. J. Bryan, **D. C. Muir**, F. Schwock, A. Yazdan-Shahmorad, R. P. N. Rao, "Robust Neural Stimulation Response Modeling Through Meta-Learning and Pretraining," *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (Under revision; arXiv:2608.26649).

RESEARCH EXPERIENCE

Undergraduate Researcher Oct 2025–Present
Neural Systems Laboratory, UW
Advisor: Dr. Rajesh Rao, Paul G. Allen School of Computer Science & Engineering

- **msTBFM: multi-session stimulation response modeling (second author, TNSRE)**
  - Co-designed and ran every experiment in the paper: Monte Carlo cross-validation, cross-animal transfer, latency benchmarks, and ablations.
  - Key results: compared with single-session models, meta-learned pretraining reduced catastrophic forecast failures from 16/40 to 1/40 sessions and calibration data needs by 50–90%.
  - Built the testing harness, analysis pipeline, statistical tests, and cloud infrastructure used to run every experiment at scale; designed most results figures.
  - Built co-adaptation, a non-meta-learning control for training and test-time adaptation, showing MAML's inner–outer loop improves sample efficiency and robustness.
- **Honors thesis: traveling-wave signals in motor cortex, toward closed-loop control**
  - Testing whether wave dynamics give a leading indicator of behavior for closed-loop stimulation; building models to forecast latent cortical dynamics under stimulation.
  - Decoding lever trajectory, trial outcome, and reaction time from generalized-phase gradients in rodent M1/M2 surface-grid recordings (Purdue collaborators).
  - Preliminary: pre-cue distributed cortical activity predicts upcoming hits vs. missed cues, surviving motor-state controls and spike-only replication.
- **Predictive model of the hippocampal–entorhinal system**
  - Extending Active Predictive Coding and the Tolman-Eichenbaum Machine into a two-stream model of the entorhinal cortex and hippocampus, with hypernetwork-generated transitions and cross-stream memory.
  - Testing for emergent grid-cell-like codes and for planning and recall in BabyAI and a custom Recall Rooms task.

Associate Researcher, VP Operations Jan 2025–Oct 2025
Sensors, Energy, and Automation Laboratory (SEAL), UW
Advisor: Dr. Alexander Mamishev, Department of Electrical & Computer Engineering

- Developed signal processing for VLF/ULF electromagnetic measurements for geophysical monitoring.
- Wrote sections of and coordinated federal and industry grant proposals (DOE STTR, NIH, NSF, CDC,

Google).
- Led a project team building lab automation and DevOps tooling.

LEADERSHIP

President; previously Vice President & New Member Educator Dec 2023–Present
Triangle Fraternity, Washington Chapter

- Grew chapter from 21 to 40 members; established its first Interfraternity Council membership and expanded the executive board from 6 to 8+ roles.
- As academic lead, chapter won the Kahlert Academic Excellence Award (highest GPA among Triangle chapters nationally).

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB, Java, Bash, Git, LaTeX
ML/AI: PyTorch, NumPy, pandas, scikit-learn, meta-learning (MAML/FOMAML), test-time adaptation, autoencoders, transformers
Neural Engineering: Neural decoding (incl. Kalman filter decoders), temporal basis function models, ECoG/cortical probe signal processing, sEMG, closed-loop BCIs
Hardware: FPGA design (SystemVerilog), analog circuits (op-amp design), EMG acquisition hardware